Room Occupancy Estimation
Milestone 1

1st Manan Mishra
Student ID: 40321141
ma_mishr@live.concordia.ca

2nd Meet Maniar
Student ID: 40039203
m_aniar@live.concordia.ca

3rd Nazmus Sakib Khan
Student ID:40309367
k_nazmu@live.concordia.ca

GitHub Repository: https://github.com/meetmaniar/INSE-6450

I. Problem Statement

The goal of our project is to build a system that estimates how many people are in a room using sensor readings. The inputs include temperature, light, sound, CO2 levels, and passive infrared (PIR) motion readings. Based on the readings collected the model will predict how many people are in the room. Because the target variable is a continuous numerical count representing the number of people in a room, this project is best described as a Regression task.

The system could be used by building managers or smart building systems to help control lighting, ventilation, heating, and energy use based on room occupancy, hence automating HVAC and lighting adjustments without any manual intervention. Also, a room can also be generalized as an enclosed indoor space hence, a well-trained and tested model can also be used to enforce safety standards in indoor arenas like concert halls and stadiums overall, helping safety officers and venue managers to get alert upon reaching a threshold capacity or take appropriate action in any case of emergency.

Good performance would mean correctly estimating occupancy levels with low average deviation from ground truth. We plan to evaluate the model using Scores like Mean Absolute Error (MAE), Root Mean Squared Error (RMSE) and Coefficient of Determination (R^2 Score).

Unlike other IOT datasets, which only capture binary states as occupied or empty, UCI's Room Occupancy Estimation dataset [1] contains labeled occupancy data along with several sensor measurements related to human presence and has no missing values. Also the sensor setup uses a star-topology edge network collecting synchronized readings across 7 specialized sensor nodes in a closed room with no HVAC systems in use.

As the Data was recorded sequentially for 4 days at the interval of 30 seconds, it is important that our splitting and sampling are done based on time-based blocks, i.e we train our model on earlier days and evaluate it on later days and not use random shuffling to avoid data leakage.

For feature engineering, it is important to note that the sensors (S1-S4) are placed all around the room recording temperatures, so it is best if we use an average of all temperatures, hence aggregating similar sensor values to get a room-wide average.

II. Data Selection

- Data Overview & Acquisition: The data set for Room occupancy Estimation was sourced from UCI Machine Learning Repository [1]. This dataset is licensed under a Creative Commons Attribution 4.0 International (CC BY 4.0). Information related to each column in the dataset can be seen in Figure 1. This dataset contains 19 columns and 10,129 records with zero missing values across all records.
  - `Date` & `Time`: Recording date and timestamp of sensor measurements. Currently, it is stored in string format and will be converted to datetime if required.
  - `S1_Temp` – `S4_Temp`: Temperature readings in degrees Celsius across 4 sensor nodes.
  - `S1_Light` – `S4_Light`: Light illuminance intensity levels in Lux across 4 sensor nodes.
  - `S1_Sound` – `S4_Sound`: Sound levels measured as raw analog voltage across 4 sensor nodes.
  - `S5_CO2`: Carbon dioxide concentration in parts per million (ppm).
  - `S5_CO2_Slope`: Rate of change in CO_2 concentration over time.
  - `S6_PIR` & `S7_PIR`: Passive infrared digital (PIR) motion detection indicators on ceiling nodes.
  - `Room_Occupancy_Count`: Ground-truth number of people present in the room (0–3).

  This dataset is appropriate for our project for room occupancy estimation because it captures multi sensor environmental changes. By combining continuously measured metrics with instantaneous motion indicators, it overcomes specific scenario-based limitations of single-sensor systems; for example, PIR sensors may fail to detect stationary occupants. Furthermore, the dataset provides precise, manually logged ground-truth counts (0 to 3 people).

```
Data columns (total 19 columns):
 #   Column               Non-Null Count  Dtype
---  ------               --------------  -----
 0   Date                 10129 non-null  str
 1   Time                 10129 non-null  str
 2   S1_Temp              10129 non-null  float64
 3   S2_Temp              10129 non-null  float64
 4   S3_Temp              10129 non-null  float64
 5   S4_Temp              10129 non-null  float64
 6   S1_Light             10129 non-null  int64
 7   S2_Light             10129 non-null  int64
 8   S3_Light             10129 non-null  int64
 9   S4_Light             10129 non-null  int64
 10  S1_Sound             10129 non-null  float64
 11  S2_Sound             10129 non-null  float64
 12  S3_Sound             10129 non-null  float64
 13  S4_Sound             10129 non-null  float64
 14  S5_CO2               10129 non-null  int64
 15  S5_CO2_Slope         10129 non-null  float64
 16  S6_PIR               10129 non-null  int64
 17  S7_PIR               10129 non-null  int64
 18  Room_Occupancy_Count  10129 non-null  int64
dtypes: float64(9), int64(8), str(2)
memory usage: 1.5 MB
```

Fig. 1. Summary of columns, data types, and non-null counts from `df.info()`.

Most of the other datasets which are more inclined towards binary classification that only tells the information about whether this room is occupied or empty where as this dataset helps us to predict the number of people in the room and hence generating a better actionable insight.

- Assumptions: For our analysis we are assuming the following things associated with the dataset:
  - Sampling & Telemetry: It is assumed that all the sensor data is collected in regular continuous intervals of 30 seconds, with the exception of a 15 day gap in the dataset, which signifies that for these days there was no data collected from any sensor.
  - Room Geometry & Sensor Setup: Throughout the time period when the data was collected we are assuming that the sensor locations and the room setup was static.
  - Closed Boundary Conditions: The Sensors were not influenced by any external factors and collection of data was done in a closed system.
  - Label Accuracy: Occupancy counts are assumed to be accurate and perfectly synchronized with the sensor data
  - Sensor Latency: Occupancy count is an instantaneous measure, whereas sensor data such as $C0_2$ might change after a certain time lag, hence the occupancy count assumes that the lag is acceptable

III. Data Quality Issues & Handling Plan

- **Missing Values:** No missing values exist in the dataset as seen in Figure 1.
- **Inconsistent Formats/Units:** The `Date` and `Time` attributes are stored as string types (`str`); these will be parsed and unified into a single continuous `datetime` feature if the project's analysis requires a time series analysis.
- **Label Noise or Weak Labels:** Ground-truth occupancy (`Room_Occupancy_Count`) was recorded manually via direct visual observation during experiments hence minimizing label noise.
- **Outliers and Erroneous Values:** Sensor noise and extreme spikes (e.g., analog sound voltage jumps or PIR triggers) can be evaluated using Box plots but cannot be discarded blindly, as extreme fluctuation is expected in a scenario where people are entering or leaving a room, Figure 2.
- **Duplicate Values:** There are no duplicate values present in the dataset, as each row has a unique timestamp.
- **Data Drift Risk (Temporal):** Sequential data recorded over 4 days might exhibits daily cyclical shifts (day vs. night); hence a temporal/chronological train-test split (rather than random shuffling) will be used to prevent data leakage and evaluate real-world generalization. It is also important to note that while the data was recorded, there was a 15 day gap starting from December 26th, 2017 to January 10th, 2018, which can be explained as a probable holiday period, Figure 3.
- **Class Imbalance:** As the room occupancy count in most cases is 0, resulting in very fewer data points of 1, 2 and ,3 it becomes crucial that before training the models the splits are done logically but not randomly. A well-balanceded training

	column	lower_bound	upper_bound	n_outliers	pct_outliers
1	S6_PIR	0.0	0.0	913	9.01
2	S7_PIR	0.0	0.0	806	7.96
3	Room_Occupancy_Count	0.0	0.0	1901	18.77
4	S2_Sound	0.04	0.07	1703	16.81
5	S5_CO2_Slope	-0.12	0.07	4033	39.82
6	S3_Sound	0.04	0.09	1671	16.5
7	S1_Sound	0.06	0.1	1772	17.49
8	S4_Sound	-0.0	0.16	721	7.12
9	S1_Temp	24.53	26.29	162	1.6
10	S2_Temp	24.53	26.29	948	9.36
11	S3_Temp	23.66	26.41	0	0.0
12	S4_Temp	24.6	26.84	0	0.0
13	S1_Light	-18.0	30.0	1716	16.94
14	S2_Light	-21.0	35.0	1017	10.04
15	S4_Light	-33.0	55.0	543	5.36
16	S3_Light	-75.0	125.0	950	9.38
17	S5_CO2	190.0	630.0	1657	16.36

Fig. 2. Outlier Number and Percentage.

```
Start Timestamp:     2017-12-22 10:49:41
End Timestamp:       2018-01-11 09:00:09
Total Duration:      19 days 22:10:28
Target Frequency:    0 days 00:00:31
Max Interval Gap:    15 days 06:15:47
```

Fig. 3. Time Stamp Information.

set will ensure that the regression model gets the data points to estimate the number of people in a room and not only the empty room, Figure 4.

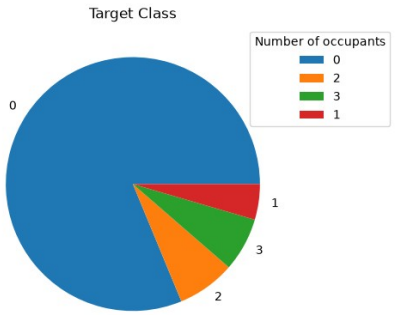


Fig. 4. Target Class.

IV. Feature Engineering

- **Candidate features:** To extract room-wide sensor information and avoid noise from any one sensor, we derive cross-node summary statistics across the spatial sensors (S_1 through S_4) for Temperature, Light, and Sound:

$$T_{\text{mean}} = \frac{S_{1,\text{Temp}} + S_{2,\text{Temp}} + S_{3,\text{Temp}} + S_{4,\text{Temp}}}{4} \tag{1}$$

$$L_{\text{mean}} = \frac{S_{1,\text{Light}} + S_{2,\text{Light}} + S_{3,\text{Light}} + S_{4,\text{Light}}}{4} \tag{2}$$

$$S_{\text{mean}} = \frac{S_{1,\text{Sound}} + S_{2,\text{Sound}} + S_{3,\text{Sound}} + S_{4,\text{Sound}}}{4} \tag{3}$$

  Also, using a heatmap, it is possible we might drop highly correlated features to remove redundancy from the dataset.
- **Feature Types:** Continuous numeric features such as T_{mean}, L_{mean}, and S_{mean} are derived. Additionally, binary categorical features is already derived from S6_PIR and S7_PIR as discrete binary indicators ($0 =$ No Motion, $1 =$ Motion Detected) to capture immediate occupant presence.

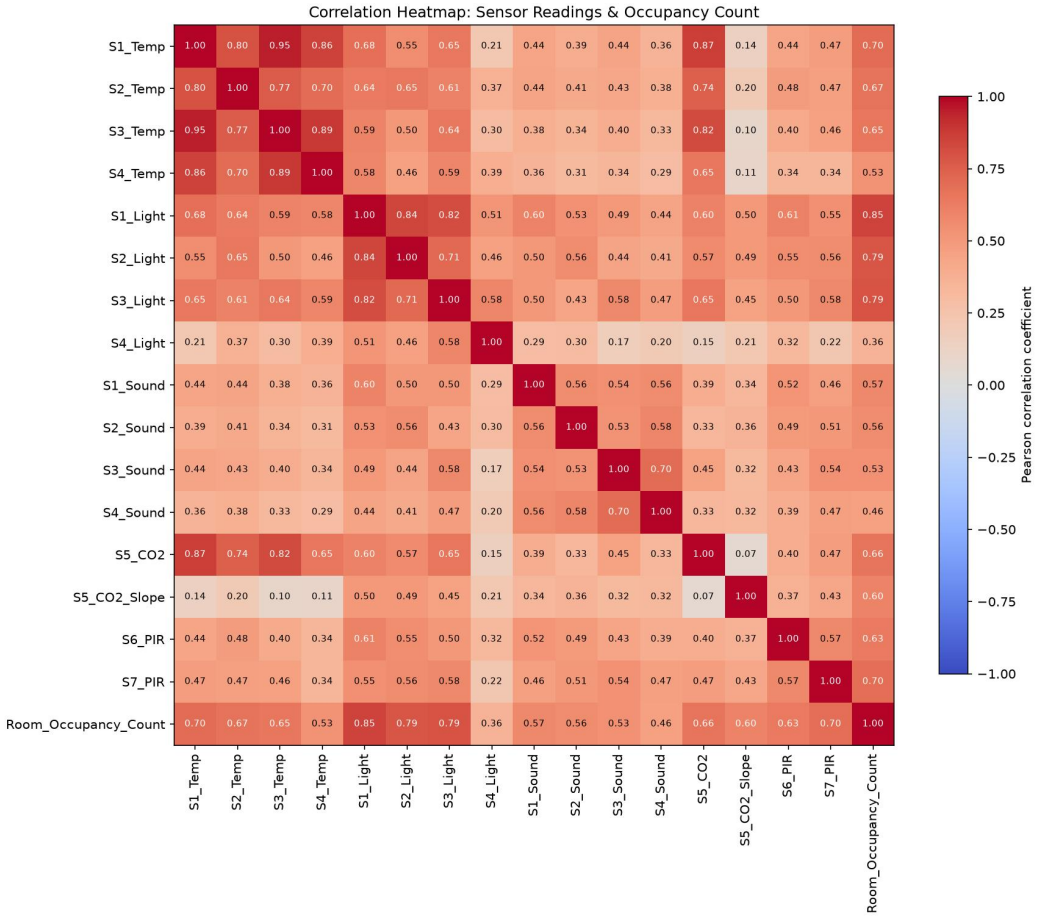


Fig. 5. Correlation Heatmap.

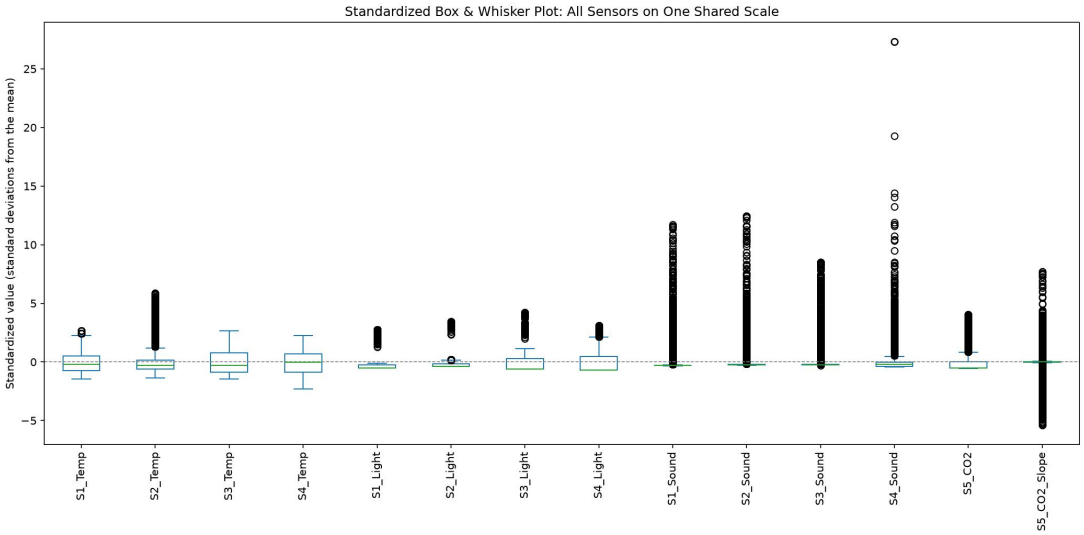


Fig. 6. Whisker Plots.

- **Domain-Specific Transformations:** As the project deals with sensor related information, it is best to use Logarithmic transformation using $\log(1+x)$ for wide-range sensor variables. This will help to compresses extreme outliers, as in our dataset it is expected to have some because if the room is empty, all sensor information will be very low, and if the room is occupied all data points are expected to increase sharply. But it is also important to note that the transformation will also depend on what models are finally implemented.
- **Categorical Encodings & Unstructured Data Handling:** The dataset consists only of continuous numerical measurements and discrete binary motion flags (PIR sensors).

References

[1] A. P. Singh and S. Chaudhari, "Room Occupancy Estimation," UCI Machine Learning Repository, 2018. [Online]. Available: https://doi.org/10.24432/C5P605